

Unit 1: Data Communication Components

Topic 1: Basics of Data Communication

1. Data Communication

Data communication is the process of exchanging data between two or more devices through a communication medium.

In simple words:

Data communication means sending data from one device to another device.

The data can be in different forms such as:

- Text
- Numbers
- Images
- Audio
- Video
- Files

For example, when you send a message from your mobile phone to another person's mobile phone, data communication takes place.

A basic data communication system contains five important components:

- Sender
- Receiver
- Message
- Transmission medium
- Protocol

2. Components of Data Communication

2.1 Sender

The sender is the device that sends the data.

The sender is also called the source.

Examples include:

- Computer
- Mobile phone
- Laptop
- Camera
- Server

Definition: Sender is the device that sends data or information to another device.

Example: If you send a message from your mobile phone, your mobile phone acts as the sender.

2.2 Receiver

The receiver is the device that receives the data.

The receiver is also called the destination.

Examples include:

- Computer
- Mobile phone
- Laptop
- Printer

- Server
- Television

Definition: Receiver is the device that receives data or information from the sender.

Example: If you send a message to your friend's mobile phone, your friend's mobile phone acts as the receiver.

2.3 Message

The message is the actual information that the sender wants to send to the receiver.

A message can contain:

- Text
- Numbers
- Images
- Audio
- Video
- Files

Definition: Message is the information or data that is communicated from the sender to the receiver.

Example: Suppose you send "Good Morning". Here, "Good Morning" is the message.

2.4 Transmission Medium

The transmission medium is the path through which the message travels from the sender to the receiver.

It is also called the communication medium or communication channel.

There are two basic types.

Wired Medium

Data travels through a physical cable.

- Twisted pair cable
- Coaxial cable
- Optical fiber

Wireless Medium

Data travels through the air using electromagnetic waves.

- Radio waves
- Microwaves
- Infrared waves

Definition: Transmission medium is the path through which data travels from the sender to the receiver.

Transmission media will be studied in greater detail later.

2.5 Protocol

A protocol is a set of rules that devices follow while communicating with each other.

In simple words:

Protocol means rules of communication.

Computers need common rules to communicate correctly. These rules are called protocols.

Examples of protocols include:

- TCP
- IP
- HTTP
- FTP
- Ethernet

Definition: A protocol is a set of rules that governs communication between two or more devices.

3. Complete Data Communication System

Suppose Computer A sends a file to Computer B.

The basic idea is: Sender sends a message through a transmission medium to the receiver according to a protocol.

4. Characteristics of Effective Data Communication

For data communication to be effective, the data should reach the correct destination, should be accurate, and should reach within the required time.

4.1 Delivery

The data must reach the correct destination.

Definition: Delivery means that data should reach the correct destination.

4.2 Accuracy

The received data should be correct and without errors. **For example,** if the sender sends COMPUTER, the receiver should receive COMPUTER and not COMUTER.

Definition: Accuracy means that the data received should be correct and without errors.

4.3 Timeliness

Data should reach the receiver within the required time. Timeliness is especially important in video calls, online gaming, live video, and voice calls.

Definition: Timeliness means that data should reach the receiver at the required time.

4.4 Jitter

Jitter means variation in the time taken by data packets to reach the receiver. It is particularly important in voice calls, video calls, online gaming, and live streaming.

Definition: Jitter is the variation in the arrival time of data packets.

5. Data Flow

Data flow means the direction in which data travels between two communicating devices.

There are three types of data flow:

- Simplex
- Half Duplex
- Full Duplex

5.1 Simplex

In simplex communication, data travels in only one direction. One device sends data and the other device only receives data.

Television broadcasting is a common example. The television station sends signals to the television. The television receives the signals but does not send the program back to the station.

Definition: Simplex is a communication mode in which data travels in only one direction.

Easy way to remember: Simplex means one way communication.

5.2 Half Duplex

In half duplex communication, data can travel in both directions, but only one direction can be used at a time. Both devices can send and receive data, but they have to take turns.

A walkie talkie is a common example. When Person A speaks, Person B listens. Then Person B can respond.

Definition: Half duplex is a communication mode in which data can travel in both directions, but not at the same time.

Easy way to remember: Half Duplex means two way communication, one direction at a time.

5.3 Full Duplex

In full duplex communication, data can travel in both directions at the same time. Both devices can send and receive data simultaneously.

A telephone or mobile phone conversation is a common example.

Definition: Full duplex is a communication mode in which data can travel in both directions at the same time.

Easy way to remember: Full Duplex means two way communication at the same time.

6. Difference Between Simplex, Half Duplex and Full Duplex

Simplex means one way.

Half Duplex means both ways, one at a time.

Full Duplex means both ways, at the same time.

7. Bandwidth

Bandwidth is an important concept in data communication.

In simple terms:

Bandwidth is the capacity of a communication channel to carry data.

A communication channel with greater capacity can carry more data.

Think about two roads. A narrow road can carry fewer vehicles. A wider road can carry more vehicles. In the same way, a communication channel with greater bandwidth can carry more data.

When bandwidth refers to the frequency range of a communication channel, it is measured in Hertz, Hz.

Definition: Bandwidth is the capacity of a communication channel to carry data.

8. Bandwidth Utilization

Bandwidth utilization means how effectively the available bandwidth of a communication channel is being used. If only a small part of the available capacity is being used, utilization is low. If most of the available capacity is being used effectively, utilization is high.

Definition: Bandwidth utilization is the effective use of the available bandwidth of a communication channel.

Bandwidth utilization is closely related to multiplexing, which allows multiple signals to share a communication channel. FDM, TDM and WDM will be studied later.

9. Complete Example

Consider a student sending a photograph from one mobile phone to another mobile phone.

Sender: The first mobile phone sends the photograph.

Message: The photograph is the message.

Transmission Medium: The network connection provides the path through which the data travels.

Protocol: Communication protocols provide the rules required for communication.

Receiver: The second mobile phone receives the photograph.

Therefore, the sender sends the message through a transmission medium to the receiver by following communication protocols.

10. Important Definitions to Learn

Data Communication: Data communication is the process of exchanging data between two or more devices through a communication medium.

Sender: Sender is the device that sends data to another device.

Receiver: Receiver is the device that receives data from another device.

Message: Message is the information or data that is communicated from the sender to the receiver.

Transmission Medium: Transmission medium is the path through which data travels from the sender to the receiver.

Protocol: A protocol is a set of rules that governs communication between two or more devices.

Simplex: Simplex is a communication mode in which data travels in only one direction.

Half Duplex: Half duplex is a communication mode in which data can travel in both directions, but not at the same time.

Full Duplex: Full duplex is a communication mode in which data can travel in both directions at the same time.

Bandwidth: Bandwidth is the capacity of a communication channel to carry data.

Bandwidth Utilization: Bandwidth utilization is the effective use of the available bandwidth of a communication channel.

Topic 2: Basics of Computer Networks

1. Introduction to Computer Networks

A computer network is a collection of two or more computers or other devices that are connected to each other so that they can communicate and share resources.

In simple words, a network allows different devices to communicate with each other.

For example, when several computers in a computer laboratory are connected together and can share files, printers, or an internet connection, they form a computer network.

A network may contain computers, mobile phones, printers, servers, routers, switches, and other network devices. These devices communicate with each other through wired or wireless communication media.

2. Why Do We Need Computer Networks?

Computer networks are used because communication and resource sharing become much easier when devices are connected.

2.1 Resource Sharing

A network allows many users to share resources. **For example,** several computers can use the same printer.

2.2 File Sharing

Users can send files from one computer to another through the network. **For example,** a teacher can share lecture notes with students over a local network.

2.3 Communication

Networks allow users to communicate using email, messaging, voice communication, and video communication.

2.4 Internet Access

Several computers can share a single internet connection through a network.

2.5 Centralized Management

In an organization, important data and services can be managed from central servers.

2.6 Remote Access

A user can access resources located on another computer or network from a different location.

3. Basic Components of a Computer Network

A basic computer network consists of devices and communication paths.

The major components are:

3.1 Nodes

A node is any device that is connected to a network and can send, receive, or process data. Examples include a computer, printer, server, router, and mobile phone. If five computers are connected to a network, each computer can be considered a node.

3.2 Link

A link is the communication path between two devices. The link may be wired or wireless. **For example,** an Ethernet cable connecting a computer to a switch provides a wired link.

3.3 Network Interface

A network interface is the point through which a device connects to a network. A network interface allows a device to send and receive data over the network. Examples include an Ethernet interface and a Wi Fi interface. A network interface is commonly associated with a network interface card, also called NIC.

4. Network Criteria

A computer network is generally evaluated using certain important criteria.

The three major criteria are:

- Performance
- Reliability
- Security

4.1 Performance

Performance tells us how well a network performs its communication tasks. It can be measured using factors such as response time, throughput, and delay.

Throughput is the amount of data successfully transferred through a network in a given amount of time. Higher throughput generally means that more data can be transferred in less time.

Response time is the amount of time taken for a request to receive a response. **For example**, when a user requests a webpage, the time taken to receive the webpage is related to response time.

4.2 Reliability

Reliability tells us how dependable a network is. A reliable network should continue working properly and should recover from failures as quickly as possible.

Reliability can be considered using frequency of failure, recovery time, and availability. A network that fails very frequently is considered less reliable.

4.3 Security

Security protects the network and its data from unauthorized access and misuse.

Network security helps protect data, devices, user information, and network services. **For example**, passwords and access control can prevent unauthorized users from accessing network resources.

5. Types of Computer Networks

Computer networks can be classified according to the geographical area covered by the network.

The major types are:

- PAN
- LAN
- MAN
- WAN

For this topic, the most important types are LAN, MAN, and WAN.

6. LAN

LAN stands for Local Area Network. A LAN is a network that covers a relatively small geographical area. Examples include a computer laboratory, office, school, college building, and home. A LAN usually provides high speed communication between connected devices.

Example: Suppose a computer laboratory contains 30 computers connected to a switch. All these computers can communicate with each other and share resources. This is an example of a LAN.

7. MAN

MAN stands for Metropolitan Area Network. A MAN covers a larger geographical area than a LAN. It can cover a city or a large part of a city. **For example**, a network connecting several offices or branches of an organization across a city can be considered a MAN.

Simple idea: LAN means building or small area. MAN means city.

8. WAN

WAN stands for Wide Area Network. A WAN covers a very large geographical area. It can connect networks across cities, states, countries, and continents. The Internet is the best known example of a very large WAN. **For example**, an organization may have offices in Delhi, Mumbai, and Bengaluru. A WAN can be used to connect the networks of these offices.

Simple idea: LAN means small area. MAN means city. WAN means large geographical area.

9. PAN

PAN stands for Personal Area Network. A PAN is a small network around an individual person. **For example**, a mobile phone connected to wireless headphones and a smartwatch forms a small personal network. PAN generally covers a very small area.

10. Comparison of LAN, MAN and WAN

11. Network Services

A network provides different services to its users and connected devices.

11.1 File Sharing

Allows users to access and exchange files over a network.

11.2 Printer Sharing

Allows multiple computers to use a common network printer.

11.3 Internet Access

Allows connected devices to access internet services.

11.4 Communication Services

Networks support services such as email, messaging, voice communication, and video communication.

11.5 Remote Access

Allows users to access resources or systems from another location.

12. Client and Server

Many computer networks use the client server model.

12.1 Client

A client is a device or program that requests a service from another computer. **For example**, a student's computer requesting a webpage from a web server acts as a client.

12.2 Server

A server is a computer or program that provides services or resources to clients. **For example**, a web server provides webpages to client computers.

Simple Example: Suppose a student opens a website. The student's computer sends a request. The web server receives the request and sends the webpage back.

Student's computer = Client

Web server = Server

13. Peer to Peer Network

In a peer to peer network, computers can communicate and share resources directly with each other.

There is no requirement for one dedicated central server for every service.

For example, two computers can directly share a file with each other.

Each computer can act as both a client and a server depending on the situation.

13.1 Advantages

- Simple to set up
- Suitable for small networks

- Does not require a dedicated server

13.2 Disadvantages

- Difficult to manage as the network becomes large
- Security management can become difficult
- Data management is not centralized

14. Client Server Network

In a client server network, one or more dedicated servers provide services to clients.

For example, in a college network, a central server may store files and provide them to different computers.

14.1 Advantages

- Centralized management
- Better control of resources
- Easier security management
- Suitable for large networks

14.2 Disadvantages

- Server failure can affect many clients
- More expensive than a simple peer to peer network
- Requires proper administration

15. Internet, Intranet and Extranet

15.1 Internet

The Internet is a worldwide network of interconnected networks. It allows devices and networks around the world to communicate.

Examples of services available through the Internet include web browsing, email, online meetings, file transfer, and online services.

15.2 Intranet

An intranet is a private network used within an organization.

For example, a college may have an internal website that can only be accessed by its staff and students. That internal network can be called an intranet.

15.3 Extranet

An extranet is a private network that allows authorized external users to access selected organizational resources.

For example, a company may allow its suppliers to access a particular part of its internal system.

16. Network Devices

A computer network requires different devices to connect computers and control data communication.

16.1 Hub

A hub connects multiple devices in a network and sends received data to all connected ports.

16.2 Switch

A switch connects devices in a network and forwards data toward the appropriate destination device.

16.3 Router

A router connects different networks and forwards packets between them.

16.4 Modem

A modem provides communication between a computer or network and a communication service by converting signals as required by the communication system.

These devices will be studied in greater detail in later topics.

17. Network Topology and Network Type

Two concepts should not be confused.

Network type generally describes the geographical area covered by a network. Examples are LAN, MAN, and WAN.

Network topology describes how devices are physically or logically arranged and connected. Examples are Bus, Star, Ring, Mesh, and Tree.

For example, a network can be a LAN in terms of its geographical coverage and use a star topology in terms of its arrangement.

We will study network topologies separately in the next topic.

18. Simple Real Life Example

Consider a college computer laboratory.

There are 40 computers connected using network cables. All computers are connected to switches. Students can share files and use a common printer. The network is limited to the college building.

In this **Example**:

Computers = Nodes

Network cables = Links

Network interfaces = Connection points of computers

Switch = Network connecting device

File sharing = Network service

College laboratory network = LAN

If the college connects its different campuses across the city, the network becomes larger and may be considered a MAN. If the organization connects offices across different cities or countries, it requires a WAN.

19. Difference Between LAN, MAN and WAN In simple words

20. Important Definitions to Learn

Computer Network: A collection of connected devices that communicate and share resources with each other.

Node: A device connected to a network that can send, receive, or process data.

Link: A communication path between two network devices.

Network Interface: The point through which a device connects to a network.

LAN: A network that covers a relatively small geographical area.

MAN: A network that covers a city or a large metropolitan area.

WAN: A network that covers a large geographical area such as multiple cities, countries, or continents.

PAN: A small network around an individual person.

Client: A device or program that requests a service.

Server: A device or program that provides a service.

Network Service: A service provided through a computer network, such as file sharing, printer sharing, or communication.

Intranet: A private network used within an organization.

Extranet: A private network that provides selected resources to authorized external users.

Topic 3: Network Topologies

1. Introduction to Network Topology

A network topology describes the way in which different devices in a computer network are arranged and connected.

In simple words, network topology tells us how the devices are connected to each other in a network.

The devices may include computers, servers, printers, switches, routers, and other network devices.

Topology can describe the physical arrangement of devices and cables, or the logical arrangement of how data moves through the network.

2. Why Is Network Topology Important?

The topology selected for a network affects how the network is installed, managed, expanded, maintained, and repaired.

For example, if one network uses a central switch and another network connects every device directly with many other devices, their structure, cost, reliability, and maintenance requirements will be different.

Therefore, choosing a suitable topology is important while designing a network.

3. Types of Network Topology

The major network topologies are:

- Bus topology
- Star topology
- Ring topology
- Mesh topology
- Tree topology
- Hybrid topology

The following internet sourced diagram shows the six major topologies covered in this topic.

Figure: Bus, Star, Ring, Mesh, Tree, and Hybrid network topologies. Internet sourced diagram.

4. Bus Topology

In a Bus topology, all devices are connected to a single main communication cable.

This main cable is called the backbone.

The data sent by one device travels along the main cable, and the intended device receives the data.

A Bus topology can be understood as a common road on which all devices are connected.

Example: Suppose five computers are connected to one common cable. All five computers use the same main communication path. This is a Bus topology.

Important point: A Bus topology normally uses a terminator at the ends of the main cable to prevent signals from reflecting back.

4.1 Advantages of Bus Topology

- It is simple to understand.
- It requires less cable than some other topologies.
- It can be relatively inexpensive for a small network.
- Adding a device can be simple in a small network.
- It does not require a central connecting device.

4.2 Disadvantages of Bus Topology

- Failure of the main cable can affect the entire network.
- Troubleshooting can be difficult.
- Network performance can decrease when many devices communicate simultaneously.
- The main cable has a limit on its length.
- Security can be difficult because data travels through the common communication path.

5. Star Topology

In a Star topology, every device is connected to a central device.

The central device is commonly a switch or hub.

All communication between devices normally passes through the central device.

Example: Suppose four computers are connected to one switch. Computer 1 sends data to Computer 3. The data first reaches the switch, and the switch forwards it toward Computer 3. This is a Star topology.

5.1 Advantages of Star Topology

- It is easy to install.
- It is easy to manage.
- Failure of one individual connection normally affects only that device.
- Fault detection is relatively easy.
- Adding or removing a device is relatively easy.
- It provides good performance when a switch is used.

5.2 Disadvantages of Star Topology

- It requires more cable than Bus topology.
- It requires a central device.
- Failure of the central device can affect the connected network.
- Installation cost can be higher than a simple Bus topology.

6. Ring Topology

In a Ring topology, each device is connected to two other devices, forming a circular arrangement.

The first device is connected to the second device, the second to the third, and so on. The last device is connected back to the first device.

Therefore, the devices form a ring.

Example: Suppose four computers are arranged in a circle. Computer 1 connects to Computer 2, Computer 2 connects to Computer 3, Computer 3 connects to Computer 4, and Computer 4 connects back to Computer 1. This forms a Ring topology.

6.1 Advantages of Ring Topology

- Data can travel in an organized manner.
- Each device gets an opportunity to participate in communication.
- Network performance can be predictable under controlled conditions.
- It can reduce the possibility of data collisions in certain ring based systems.

6.2 Disadvantages of Ring Topology

- Failure of a connection can affect communication around the ring.
- Adding or removing devices can disturb the network.
- Troubleshooting can be difficult.
- Maintenance can be more difficult than Star topology.

7. Mesh Topology

In a Mesh topology, devices are connected to multiple other devices.

There are two major forms: Full Mesh and Partial Mesh.

7.1 Full Mesh Topology

In a Full Mesh topology, every device is directly connected to every other device.

Suppose there are four devices: A, B, C, and D. Each device has a direct connection with all the other devices.

Therefore, A connects with B, C, and D. B connects with A, C, and D. C connects with A, B, and D. D connects with A, B, and C.

7.2 Partial Mesh Topology

In a Partial Mesh topology, only some devices have direct connections with multiple other devices.

Therefore, it requires fewer connections than a Full Mesh topology.

7.3 Advantages of Mesh Topology

- It provides high reliability.
- Multiple communication paths are available.
- Failure of one connection does not necessarily stop communication between other devices.

- It provides good fault tolerance.
- It can be useful where reliability is very important.

7.4 Disadvantages of Mesh Topology

- It requires a large number of connections.
- Installation can be expensive.
- Configuration can be complicated.
- Maintenance can be difficult.
- Full Mesh becomes impractical as the number of devices increases.

8. Tree Topology

A Tree topology is a hierarchical topology.

It combines characteristics of Star and Bus topologies.

Devices are arranged in different levels.

A higher level device can connect to several lower level devices.

Example: Consider a college network. One main network device connects to several department level devices. Each department level device then connects to computers in that department. This creates a hierarchical structure similar to a tree.

8.1 Advantages of Tree Topology

- It supports hierarchical network organization.
- It is suitable for larger networks.
- Different sections of the network can be managed separately.
- New sections can be added relatively easily.
- It provides a structured network design.

8.2 Disadvantages of Tree Topology

- It requires more cabling than simple topologies.
- It can be more expensive.
- Failure of an important higher level device can affect a large part of the network.
- Configuration and maintenance can become complex.

9. Hybrid Topology

A Hybrid topology is formed by combining two or more different network topologies.

For example, an organization may use Star topology with Bus topology, or Star topology with Ring topology. Such a network is called a Hybrid topology.

Hybrid topology is useful when different parts of an organization have different networking requirements.

9.1 Advantages of Hybrid Topology

- It provides flexibility.
- Different topologies can be used for different network sections.
- It can be designed according to organizational requirements.
- It can support large and complex networks.
- Failure in one section may not necessarily affect the entire network.

9.2 Disadvantages of Hybrid Topology

- Design can be complicated.
- Installation can be expensive.
- Maintenance can be difficult.
- Network management requires proper planning.

10. Comparison of Different Network Topologies

11. Performance Parameters Used to Compare Network Topologies

11.1 Cost

Cost includes the expense of cables, network devices, installation, and maintenance. A topology requiring many connections generally has a higher installation cost.

11.2 Performance

Performance describes how efficiently data can be transmitted through the network. Factors such as traffic, delay, throughput, and number of devices can affect performance.

11.3 Reliability

Reliability tells us how dependable the network is. A topology with multiple alternative paths can continue communication even when one path fails.

11.4 Fault Tolerance

Fault tolerance refers to the ability of a network to continue operating when some component fails. A network with alternative communication paths generally has better fault tolerance.

11.5 Scalability

Scalability refers to how easily a network can be expanded. **For example**, adding a new computer to a Star topology is generally easier than adding a new computer to a Full Mesh topology.

11.6 Maintenance

Maintenance refers to how easily the network can be monitored, repaired, and managed. A topology with simpler connections is generally easier to maintain.

11.7 Installation Complexity

Some topologies are easy to install, while others require many connections and careful planning. Full Mesh topology has high installation complexity because many direct connections are required.

12. Physical Topology and Logical Topology

12.1 Physical Topology

Physical topology describes the actual physical arrangement of devices and cables.

It answers the question: How are the devices physically connected?

For example, computers connected to a central switch using cables form a physical Star arrangement.

12.2 Logical Topology

Logical topology describes how data actually flows between devices in a network.

It answers the question: How does data move through the network?

Physical and logical topology do not always have to be the same.

13. How to Choose a Network Topology

13.1 Number of Devices

A topology suitable for ten devices may not be suitable for thousands of devices.

13.2 Cost

The available budget affects the choice of cables, devices, and network structure.

13.3 Reliability

If network availability is very important, a topology with multiple paths may be preferred.

13.4 Expansion

If more devices will be added in the future, a topology that supports easy expansion is useful.

13.5 Maintenance

A topology should ideally be easy to troubleshoot and maintain.

13.6 Performance

The expected amount of network traffic should also be considered.

14. Simple Real Life Example

- Consider a college.
- There are several departments: Computer Science, Mechanical Engineering, Electrical Engineering, and Civil Engineering.
- Each department has several computers.
- A switch can be used to connect computers within each department.
- The department switches can then be connected to a higher level network device.
- This creates a hierarchical structure.
- Therefore, the overall network can use a Tree topology.
- If different sections use different topologies and these sections are connected together, the resulting network can be a Hybrid topology.

15. Quick Understanding of All Topologies

16. Important Definitions to Learn

Network Topology: The arrangement or structure of devices and connections in a computer network.

Bus Topology: A topology in which all devices share a common main communication cable.

Star Topology: A topology in which all devices are connected to a central device.

Ring Topology: A topology in which each device is connected to two other devices, forming a circular structure.

Mesh Topology: A topology in which devices are connected to multiple other devices to provide multiple communication paths.

Full Mesh: A mesh topology in which every device has a direct connection with every other device.

Partial Mesh: A mesh topology in which only some devices have multiple direct connections.

Tree Topology: A hierarchical topology in which devices are arranged in different levels.

Hybrid Topology: A topology formed by combining two or more different topologies.

Physical Topology: The physical arrangement of devices and communication media.

Logical Topology: The way data logically flows between devices in a network.

Scalability: The ability of a network to expand when more devices or users are added.

Fault Tolerance: The ability of a network to continue functioning when some components fail.

Component	Example
Sender	Computer A
Message	File
Transmission Medium	Network cable or Wi Fi
Receiver	Computer B
Protocol	Rules followed during communication

Feature	Simplex	Half Duplex	Full Duplex
Direction	One direction	Both directions	Both directions
Same time communication	No	No	Yes
Communication type	One way	Two way, one at a time	Two way, at the same time
Example	Television broadcasting	Walkie talkie	Telephone

Feature	LAN	MAN	WAN
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Full form	Local Area Network	Metropolitan Area Network	Wide Area Network
Coverage	Small area	City or large campus	Very large geographical area
Example	Computer laboratory	City wide organization network	Internet
Speed	Generally high	Generally lower than LAN	Generally lower than LAN
Ownership	Usually private	Private or public	Often uses multiple organizations and service providers
Purpose	Connect nearby devices	Connect networks across a city	Connect networks over large distances

Network	Simple Meaning
PAN	Network around one person
LAN	Network in a small area
MAN	Network covering a city
WAN	Network covering a very large area

Feature	Bus	Star	Ring	Mesh	Tree	Hybrid
Basic structure	Common backbone	Central device	Circular	Multiple connections	Hierarchical	Combination
Central device required	No	Yes	Usually no	No	Usually at higher levels	Depends on design
Cost	Low	Moderate	Moderate	High	Moderate to high	Depends on design
Installation	Easy	Easy	Moderate	Difficult	Moderate	Difficult
Reliability	Low	Good	Moderate	Very high	Good	Depends on design
Fault detection	Difficult	Easy	Moderate	Relatively easy	Moderate	Can be difficult
Expansion	Moderate	Easy	Difficult	Difficult	Easy for sections	Flexible

Topic 4: Protocols and Standards

1. Introduction to Protocols

When two or more devices communicate with each other, they need a common set of rules. These rules tell the devices how data should be prepared, sent, received, understood, and processed. These rules are called protocols. A protocol is a set of rules that governs communication between two or more devices in a network.

2. Why Are Protocols Needed?

Protocols are needed because network devices must follow the same rules to communicate correctly.

The main purposes of protocols are:

- To identify how communication should begin and end.
- To define the format of data.
- To define how data is transmitted.
- To control the flow of data between devices.
- To detect and handle errors.

- To identify the sender and receiver.
- To provide reliable communication when required.

3. Elements of a Protocol

A protocol mainly defines three important elements.

3.1 Syntax

Syntax defines the structure or format of the data. It tells us how the information is arranged.

3.2 Semantics

Semantics defines the meaning of each section of the data. It tells the devices what the information represents and what action should be taken.

3.3 Timing

Timing defines when data should be sent and how fast it should be transmitted. It also helps control the rate of communication.

4. Simple Example of a Protocol

Consider two students communicating through a messaging application. Both students must use the same communication rules so that the message can be delivered and understood.

For example, the communication system must know who is sending the message, who should receive it, how the message is represented, and when the information should be transmitted.

In a computer network, protocols perform the same basic role, but the rules are much more precise and are followed automatically by network devices and software.

5. Protocols Used in Computer Networks

Different protocols are used for different communication tasks. Some common protocols are given below.

Protocol	Main Purpose
HTTP	Used for communication between web browsers and web servers.
HTTPS	Used for secure communication between web browsers and web servers.
FTP	Used for transferring files between computers.
SMTP	Used for sending email messages.
POP3	Used for receiving email messages.
IMAP	Used for accessing and managing email messages on a mail server.
DNS	Used to translate domain names into IP addresses.
TCP	Provides reliable and ordered delivery of data.
UDP	Provides fast communication without the same reliability mechanisms as TCP.
IP	Provides addressing and routing of packets across networks.

6. Protocols at Different Layers

Network communication is divided into layers. Different protocols perform different tasks at different layers.

Layer or Area	Examples of Protocols	Main Purpose
Application	HTTP, HTTPS, FTP, SMTP, DNS	Provides services used by network applications.
Transport	TCP, UDP	Provides end to end data delivery between applications.
Network	IP	Provides logical addressing and routing.
Data Link	Ethernet, PPP	Controls communication over a local link.

7. What Is a Standard?

A standard is a set of agreed rules, specifications, or guidelines that allows different devices, systems, and software to work together.

Standards are important because computer networks are built using equipment and software from many different manufacturers.

8. Why Are Standards Needed?

Standards provide a common way of designing and operating network technologies.

- They allow devices from different manufacturers to communicate.
- They improve compatibility between hardware and software.
- They make network design easier.
- They reduce dependence on a single manufacturer.
- They support large scale development of network technologies.
- They provide common technical specifications.

9. Types of Standards

9.1 De Facto Standard

A de facto standard becomes widely accepted because many people, companies, or organizations use it. It may become a standard through widespread use rather than through a formal approval process.

9.2 De Jure Standard

A de jure standard is formally approved by a recognized standards organization.

10. Organizations Related to Network Standards

Several organizations develop, publish, or maintain standards used in computer networking.

Organization	Main Role
ISO	Develops international standards. It developed the OSI reference model.
IEEE	Develops standards for many networking technologies, including Ethernet and wireless LANs.
IETF	Develops and publishes Internet standards and technical documents, including standards related to TCP and IP.
ITU T	Develops international telecommunications standards.
ANSI	Coordinates and approves standards in the United States and works with international standardization activities.

11. Protocol and Standard

A protocol defines rules for communication between devices. A standard defines agreed specifications that help different systems, devices, and technologies work together.

Basis	Protocol	Standard
Meaning	Rules used for communication.	Agreed specifications or guidelines.
Purpose	Controls how communication takes place.	Ensures compatibility and common implementation.
Example	TCP, IP, HTTP.	IEEE 802.3, IEEE 802.11.
Main Focus	Communication behavior.	Common technical specifications.

12. Protocols and Standards Work Together

Protocols and standards are closely related. A standard can define the technical requirements for a networking technology, while protocols define how communication takes place within that technology.

For example, Ethernet is standardized by IEEE 802.3. Wireless LAN technology is standardized by the IEEE 802.11 family. These standards help devices from different manufacturers communicate using common technical rules.

13. Protocol Suite

A protocol suite is a collection of related protocols that work together to provide complete network communication.

The TCP IP protocol suite is a well known example. It contains protocols that perform different communication tasks, including addressing, routing, reliable delivery, and application communication.

14. Rules Defined by Network Protocols

A network protocol may define rules related to:

- Data format
- Addressing
- Transmission method
- Error handling
- Flow control
- Connection establishment
- Connection termination
- Data ordering
- Security and authentication when supported

15. Simple Real Life Example

Think about traffic rules on a road. Vehicles need common rules about lanes, signals, speed, and right of way. Without common rules, communication between drivers would become difficult and unsafe.

Similarly, protocols provide communication rules for network devices. Standards provide common technical specifications so that devices made by different manufacturers can work together.

16. Important Definitions to Learn

Protocol: A set of rules that governs communication between two or more devices.

Syntax: The structure or format of data used during communication.

Semantics: The meaning of different parts of data and the actions associated with them.

Timing: The rules related to when data is sent and the rate of transmission.

Standard: An agreed set of specifications or guidelines used to provide compatibility between systems.

De Facto Standard: A standard that becomes widely accepted through common use.

De Jure Standard: A standard that is formally approved by a recognized standards organization.

Protocol Suite: A collection of related protocols that work together for network communication.

Topic 5: OSI Model

1. Introduction to the OSI Model

The OSI model is a reference model used to understand how communication takes place between computers over a network.

OSI stands for Open Systems Interconnection. The model was developed by the International Organization for Standardization, commonly known as ISO.

The OSI model divides network communication into seven separate layers. Each layer performs a specific set of functions and works with the layers above and below it.

2. Why Was the OSI Model Developed?

Different computer manufacturers originally used different communication methods. This made communication between different systems difficult.

The OSI model provides a common framework for understanding network communication. It divides the complete communication process into smaller and easier to manage functions.

The main purposes of the OSI model are:

- To divide network communication into manageable layers.
- To provide a standard framework for network communication.
- To help different systems communicate using common concepts.
- To make network design and troubleshooting easier.
- To allow changes in one layer without requiring complete changes in other layers.

3. Seven Layers of the OSI Model

Layer Number	Layer Name	Main Function
7	Application	Provides network services to applications used by the user.
6	Presentation	Handles translation, encryption, and compression of data.
5	Session	Establishes, manages, and terminates communication sessions.
4	Transport	Provides end to end delivery, flow control, and error control.
3	Network	Provides logical addressing and routing.
2	Data Link	Provides framing, physical addressing, and error detection on a link.
1	Physical	Transmits raw bits through the physical medium.

4. Layer 1: Physical Layer

The Physical Layer is the lowest layer of the OSI model. It is responsible for transmitting raw bits from one device to another through a physical medium.

Main functions of the Physical Layer include:

- Transmission of bits.
- Definition of physical connections.
- Representation of bits as electrical, optical, or radio signals.
- Definition of transmission rate.
- Definition of physical characteristics of interfaces and media.

Examples of devices associated with the Physical Layer include hubs, repeaters, cables, and connectors.

5. Layer 2: Data Link Layer

The Data Link Layer provides reliable communication between directly connected devices over a physical link.

Main functions of the Data Link Layer include:

- Framing.
- Physical addressing using MAC addresses.
- Error detection.

- Flow control in supported technologies.
- Controlling access to a shared transmission medium.

Examples include Ethernet and other data link technologies.

6. Layer 3: Network Layer

The Network Layer is responsible for delivering packets from the source network to the destination network.

Main functions of the Network Layer include:

- Logical addressing.
- Routing.
- Packet forwarding.
- Selection of paths between networks.

The Internet Protocol, or IP, operates at the Network Layer.

7. Layer 4: Transport Layer

The Transport Layer provides end to end communication between applications running on different devices.

Main functions of the Transport Layer include:

- Segmentation and reassembly of data.
- End to end delivery.
- Flow control.
- Error control.
- Connection management when required.
- Identification of applications using port numbers.

TCP and UDP are important Transport Layer protocols.

8. Layer 5: Session Layer

The Session Layer establishes, manages, and terminates communication sessions between applications.

Its functions include:

- Establishing a session.
- Maintaining a session.
- Managing communication between applications.
- Terminating a session.
- Providing synchronization points when required.

9. Layer 6: Presentation Layer

The Presentation Layer is concerned with the representation and format of data so that different systems can understand the information.

Its important functions include:

- Data translation.
- Data encryption and decryption.
- Data compression and decompression.
- Conversion between different data representation formats.

10. Layer 7: Application Layer

The Application Layer is the top layer of the OSI model. It provides network related services to applications used by the user.

Its functions include:

- Providing services to network applications.
- Supporting services such as web access, file transfer, and email communication.

- Allowing applications to access network services.

Examples of application layer protocols include HTTP, FTP, SMTP, and DNS.

11. Easy Way to Understand the Seven Layers

A simple way to remember the purpose of the layers is to think about sending a message from one computer to another.

Layer	Easy Meaning
Application	Provides network services to applications.
Presentation	Makes data understandable and handles format, encryption, and compression.
Session	Manages the communication session.
Transport	Provides end to end delivery.
Network	Finds and manages the path between networks.
Data Link	Moves frames between directly connected devices.
Physical	Transmits bits as signals.

12. Communication Between OSI Layers

Each OSI layer provides services to the layer above it and uses services provided by the layer below it.

When data is sent, it moves from the Application Layer down through the seven layers. Each layer adds information required for its function. At the receiving device, the process takes place in the reverse direction.

13. Encapsulation

Encapsulation is the process of adding control information to data as it moves down through the OSI layers at the sender side.

For example, the Transport Layer may divide application data into segments. The Network Layer adds information needed for packet delivery. The Data Link Layer adds information needed for frame delivery.

14. Decapsulation

Decapsulation is the process of removing the control information added by the different layers as data moves upward through the OSI layers at the receiver side.

The receiver processes the information layer by layer until the original data is delivered to the destination application.

15. Data Units at Different Layers

OSI Layer	Common Data Unit
Application	Data
Presentation	Data
Session	Data
Transport	Segment for TCP and Datagram for UDP
Network	Packet
Data Link	Frame
Physical	Bits

16. Addressing in the OSI Model

Different layers use different types of addressing or identification information for communication.

Layer	Address or Identification	Purpose
Transport	Port Number	Identifies the application or service.
Network	IP Address	Identifies the source and destination devices or interfaces across networks.

Data Link	MAC Address	Identifies devices or interfaces on a local network.
Physical	Signals and physical connections	Carries bits through the physical medium.

17. OSI Model and Network Devices

Device	Commonly Associated Layer
Repeater	Physical Layer
Hub	Physical Layer
Bridge	Data Link Layer
Switch	Data Link Layer
Router	Network Layer

Some modern network devices can perform functions at more than one layer. Therefore, the table shows the common layer associated with the basic function of each device.

18. Advantages of the OSI Model

- It divides complex network communication into smaller layers.
- It makes learning and understanding networking easier.
- It helps in troubleshooting network problems.
- It provides a common reference framework.
- Changes in one layer can often be handled independently of other layers.
- It helps in designing and comparing networking technologies.

19. Limitations of the OSI Model

- It is mainly a reference model rather than a complete protocol implementation.
- Some real networking protocols do not fit perfectly into only one OSI layer.
- Some functions are divided differently in practical protocol suites.
- The seven layer structure can sometimes make practical networking concepts appear more complex.

20. OSI Model and TCP IP Model

The TCP IP model is the protocol architecture used for the Internet. The OSI model is mainly a reference model used to understand network communication.

OSI Model	TCP IP Model
7 layers	Commonly represented using 4 layers
Application, Presentation, Session are separate layers.	These functions are generally combined into the Application layer.
Transport layer	Transport layer
Network layer	Internet layer
Data Link and Physical layers are separate.	These functions are commonly represented together in the Network Access or Link layer.

21. Difference Between OSI and TCP IP

Basis	OSI Model	TCP IP Model
Type	Reference model.	Protocol architecture and suite.
Number of Layers	Seven.	Commonly four.
Development	Developed by ISO.	Developed through the Internet protocol development process.
Use	Useful for understanding and teaching network communication.	Widely used in practical Internet communication.
Layer Structure	Has separate Session and Presentation layers.	Combines their functions with the Application layer.

22. Simple Real Life Example

Imagine sending a parcel from one city to another. First, the sender prepares the contents. The parcel is packed, labelled, assigned to a route, transported between locations, and finally delivered to the receiver.

Similarly, the OSI model divides network communication into separate functions. Each layer performs its own task and passes the result to the next layer.

23. Important Definitions to Learn

OSI Model: A seven layer reference model that describes how data communication takes place between networked systems.

Physical Layer: The lowest OSI layer responsible for transmitting raw bits through a physical medium.

Data Link Layer: The layer responsible for framing, physical addressing, and communication over a local link.

Network Layer: The layer responsible for logical addressing and routing packets between networks.

Transport Layer: The layer responsible for end to end delivery and transport services between applications.

Session Layer: The layer responsible for establishing, managing, and terminating communication sessions.

Presentation Layer: The layer responsible for data translation, encryption, and compression.

Application Layer: The top OSI layer that provides network services to applications.

Encapsulation: The process of adding control information as data moves down the layers at the sender.

Decapsulation: The process of removing control information as data moves up the layers at the receiver.

Topic 6: Transmission Media

1. Introduction to Transmission Media

Transmission media is the path through which data travels from a sender to a receiver in a communication system. It is also called a communication medium or communication channel. The choice of transmission medium affects speed, distance, cost, reliability, and security of communication.

2. Types of Transmission Media

Transmission media are mainly divided into two categories:

Type	Meaning	Examples
Guided Media	Data travels through a physical path.	Twisted pair cable, coaxial cable, optical fiber
Unguided Media	Data travels through air or free space using electromagnetic waves.	Radio waves, microwaves, infrared

3. Guided Transmission Media

Guided media use a physical cable or wire to carry signals from one device to another.

The major guided transmission media used in computer networks are twisted pair cable, coaxial cable, and optical fiber.

4. Twisted Pair Cable

Twisted pair cable consists of two insulated copper wires twisted around each other. The twisting helps reduce interference from nearby wires and external electrical sources.

Twisted pair cable is widely used in telephone systems and local area networks.

4.1 Types of Twisted Pair Cable

Type	Description
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UTP	Unshielded Twisted Pair does not have an additional metallic shield around the twisted pairs.
STP	Shielded Twisted Pair has additional shielding that provides better protection against electromagnetic interference.

4.2 Advantages of Twisted Pair Cable

1. Low cost.
2. Easy to install.
3. Easy to maintain.
4. Suitable for local area networks.
5. Available in many categories with different performance levels.

4.3 Disadvantages of Twisted Pair Cable

1. More affected by interference than optical fiber.
2. Limited transmission distance compared with optical fiber.
3. Lower bandwidth than optical fiber.
4. Signal quality can decrease over long distances.

5. Coaxial Cable

Coaxial cable contains a central conductor surrounded by insulation, a metallic shield, and an outer protective covering.

The metallic shield helps protect the signal from external interference. Coaxial cable has been used in cable television systems, broadband communication, and some older network technologies.

5.1 Advantages of Coaxial Cable

1. Better protection from interference than ordinary copper wire.
2. Higher bandwidth than basic twisted pair in many applications.
3. Strong physical construction.
4. Suitable for longer distances than some basic copper connections.

5.2 Disadvantages of Coaxial Cable

1. More expensive than basic twisted pair.
2. More difficult to install than twisted pair.
3. Heavier and less flexible than twisted pair.

6. Optical Fiber

Optical fiber is a transmission medium that carries data in the form of light signals through very thin strands of glass or plastic.

An optical fiber normally contains a core through which light travels, a cladding around the core, and a protective outer covering.

6.1 Working of Optical Fiber

The sender converts electrical data into light signals. The light travels through the fiber and reaches the receiver.

The receiver converts the light signals back into electrical data.

The light is guided through the core by the principle of total internal reflection.

6.2 Types of Optical Fiber

Type	Description
Single Mode Fiber	Uses a very small core and mainly carries one mode of light. It supports very long distances and high data rates.

Multimode Fiber	Uses a larger core and allows multiple modes of light to travel through the fiber. It is commonly used for shorter distances.
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6.3 Advantages of Optical Fiber

1. Very high bandwidth.
2. Supports long distance communication.
3. Very low signal loss compared with many copper media.
4. Not affected by electromagnetic interference.
5. Provides good security because the signal is difficult to intercept without disturbing the fiber.
6. Lightweight compared with many copper cable systems.

6.4 Disadvantages of Optical Fiber

1. Higher installation cost in many situations.
2. Installation and repair require specialized skills and equipment.
3. Fiber can be physically damaged if handled improperly.
4. Connecting and joining fibers requires greater care.

7. Comparison of Guided Media

Feature	Twisted Pair	Coaxial Cable	Optical Fiber
Medium	Copper	Copper	Glass or plastic
Signal	Electrical	Electrical	Light
Bandwidth	Moderate	Higher than basic twisted pair in many uses	Very high
Distance	Short to moderate	Moderate	Long
Interference	More susceptible	Better protection	Very low susceptibility
Cost	Low	Moderate	Higher
Installation	Easy	Moderate	More difficult

8. Unguided Transmission Media

Unguided media do not use a physical cable. Data is transmitted through air or free space using electromagnetic waves.

Unguided media are also called wireless transmission media.

9. Radio Waves

Radio waves are electromagnetic waves that can travel through air and can often cover large areas.

They are widely used in wireless communication systems such as radio communication, mobile communication, and wireless networks.

9.1 Advantages of Radio Waves

1. Do not require a physical cable.
2. Can cover large areas.
3. Useful for mobile communication.
4. Can pass through some obstacles depending on frequency and environment.

9.2 Disadvantages of Radio Waves

1. Can suffer from interference.
2. Security can be more difficult because signals propagate through open space.
3. Performance can be affected by the environment.

10. Microwaves

Microwaves are high frequency electromagnetic waves used for wireless communication. They are commonly used for point to point communication and satellite communication.

Terrestrial microwave links generally require a clear path between the transmitting and receiving antennas.

10.1 Advantages of Microwaves

1. Suitable for long distance wireless communication.
2. High data carrying capacity.
3. Useful where laying cables is difficult.
4. Can be used for communication between distant locations.

10.2 Disadvantages of Microwaves

1. Usually require line of sight.
2. Buildings, hills, and other obstacles can block the signal.
3. Weather conditions can affect some microwave communication systems.
4. Accurate antenna alignment may be required.

11. Infrared

Infrared communication uses infrared electromagnetic waves to transmit data over short distances.

Infrared signals generally do not pass through walls, which limits communication to a room or a nearby area in many applications.

11.1 Advantages of Infrared

1. Useful for short distance communication.
2. Does not require a physical cable.
3. Does not normally pass through walls, which can reduce interference between nearby rooms.

11.2 Disadvantages of Infrared

1. Short transmission range.
2. Obstacles can block the signal.
3. Not suitable for communication through walls.
4. Strong sunlight and other infrared sources can affect communication.

12. Comparison of Unguided Media

Feature	Radio Waves	Microwaves	Infrared
Transmission	Wireless	Wireless	Wireless
Typical Range	Short to very large depending on system	Long distance	Short distance
Line of Sight	Not always required	Generally required for terrestrial links	Usually required
Common Uses	Mobile communication, wireless networks	Satellite and point to point links	Short range device communication
Interference	Can be affected by interference	Can be affected by weather and obstacles	Can be affected by environmental infrared sources

13. Performance Parameters of Transmission Media

Transmission media can be compared using several performance parameters.

13.1 Bandwidth

Bandwidth represents the capacity of a communication channel to carry data. A higher bandwidth generally allows more data to be transmitted in a given period.

13.2 Transmission Distance

Transmission distance is the distance over which a signal can travel while maintaining acceptable communication quality.

13.3 Attenuation

Attenuation is the loss of signal strength as a signal travels through a transmission medium.

Greater attenuation means that the signal becomes weaker more quickly. Repeaters or amplifiers may be required in some systems to maintain signal quality.

13.4 Distortion

Distortion occurs when the shape or form of a signal changes while it travels through the transmission medium.

13.5 Noise

Noise is an unwanted signal that can interfere with the original data signal. Noise can cause errors in the received data.

13.6 Interference

Interference occurs when unwanted electromagnetic signals affect the communication signal. Guided and unguided media can experience different forms of interference.

13.7 Latency

Latency is the time taken for data or a signal to travel from the sender toward the receiver.

13.8 Reliability

Reliability indicates how consistently a transmission medium can carry data correctly without frequent failures or errors.

14. Attenuation and Latency

Term	Meaning	Effect
Attenuation	Loss of signal strength during transmission.	Weakens the received signal.
Latency	Time taken for data or a signal to travel.	Causes delay in communication.

15. Wired and Wireless Transmission

Basis	Wired	Wireless
Medium	Uses a physical cable.	Uses air or free space.
Examples	Twisted pair, coaxial cable, optical fiber.	Radio waves, microwaves, infrared.
Mobility	Limited by cable connection.	Supports greater mobility.
Installation	Requires cable installation.	Usually avoids physical cable installation.
Security	Physical access is generally required to access the cable.	Signals propagate through open space and require suitable security measures.
Interference	Depends on the cable type.	Can be more affected by environmental interference.

16. Factors for Selecting Transmission Media

1. Required bandwidth.
2. Required transmission distance.
3. Installation cost.
4. Maintenance cost.
5. Level of interference in the environment.
6. Required reliability.

7. Security requirements.
8. Mobility requirements.
9. Physical installation conditions.

17. Simple Real Life Example

Suppose a college needs to connect computers in a computer laboratory. Twisted pair cable may be suitable because it is affordable and easy to install.

If the college needs to connect buildings over a long distance with very high data capacity, optical fiber may be more suitable.

If students need to connect mobile devices without cables, a wireless medium such as radio waves may be used.

18. Important Definitions to Learn

Transmission Media: The path through which data travels from a sender to a receiver.

Guided Media: A transmission medium in which signals travel through a physical path such as a cable.

Unguided Media: A transmission medium in which signals travel through air or free space.

Twisted Pair Cable: A cable made of two insulated copper wires twisted around each other.

Coaxial Cable: A cable containing a central conductor, insulation, metallic shielding, and an outer covering.

Optical Fiber: A transmission medium that carries data as light signals through glass or plastic fiber.

Radio Waves: Electromagnetic waves used for wireless communication over different ranges.

Microwaves: High frequency electromagnetic waves used for wireless communication, including point to point and satellite links.

Infrared: Electromagnetic radiation used for short distance wireless communication.

Attenuation: Loss of signal strength as a signal travels through a transmission medium.

Distortion: Change in the shape or form of a signal during transmission.

Noise: An unwanted signal that interferes with the original data signal.

Latency: The time taken for data or a signal to travel from one point toward another.

Topic 7: LAN, Wired LAN and Wireless LAN

1. Introduction to LAN

LAN stands for Local Area Network. A LAN is a computer network that connects devices within a limited geographical area such as a room, laboratory, office, building, or college campus.

LANs are commonly used to share resources such as files, printers, applications, and Internet connections between connected devices.

2. Main Characteristics of a LAN

1. Covers a relatively small geographical area.
2. Usually provides high data transfer speed.
3. Is generally owned and managed by a single organization or user.
4. Allows connected devices to share resources.
5. Can use wired or wireless transmission.

3. Components of a LAN

Component	Purpose
Computers and Devices	Act as hosts that send, receive, and process data.
Network Interface	Provides the connection between a device and the network.
Switch	Connects devices in a LAN and forwards frames toward the appropriate destination.

Router	Connects the LAN to other networks.
Transmission Medium	Carries data between devices. It may be wired or wireless.
Access Point	Provides wireless connectivity to devices in a wireless LAN.

4. Wired LAN

A wired LAN connects devices using physical transmission media. Ethernet is the most widely used technology for wired LANs.

In a typical wired LAN, computers and other devices are connected to a network switch using Ethernet cables.

5. Ethernet

Ethernet is a widely used LAN technology. Ethernet standards are developed by the IEEE under the IEEE 802.3 family.

Ethernet can operate at different data rates and can use different physical media. Modern Ethernet networks commonly use twisted pair cables or optical fiber.

6. Working of a Wired LAN

1. A device prepares data for transmission.
2. The network interface converts the data into signals suitable for the transmission medium.
3. The switch receives the Ethernet frame.
4. The switch examines the destination MAC address and forwards the frame toward the appropriate port.
5. The destination device receives and processes the frame.

7. Advantages of Wired LAN

1. Provides reliable communication.
2. Can provide high data transfer rates.
3. Physical connections can provide predictable performance.
4. Suitable for offices, laboratories, and other fixed locations.
5. Can provide strong network security when physical access is controlled.

8. Disadvantages of Wired LAN

1. Requires physical cables.
2. Installation may require additional wiring work.
3. Devices have limited mobility.
4. Cable damage can affect connectivity.

9. Wireless LAN

Wireless LAN, commonly called WLAN, is a LAN in which devices communicate without physical network cables. Wireless communication generally uses radio waves.

The IEEE 802.11 family of standards is used for wireless LAN technology.

10. IEEE 802.11

IEEE 802.11 is a family of standards for wireless local area networks. Different versions of the standard provide different capabilities, frequencies, channel widths, and data rates.

A wireless access point is commonly used to connect wireless devices to the LAN.

11. Working of a Wireless LAN

1. A wireless device such as a laptop or smartphone prepares data for transmission.
2. The wireless network interface converts the data into a radio signal.
3. The signal is transmitted through the air.

4. A wireless access point receives the transmission and communicates with the wired or wireless network.
5. The destination device receives the data.

12. Access Point

An access point, or AP, is a networking device that provides wireless connectivity to client devices and connects the wireless network to the rest of the network.

For example, in a college campus, an access point can allow students to connect laptops and smartphones to the campus network using Wi Fi.

13. Wi Fi

Wi Fi is a technology used to provide wireless network connectivity based on IEEE 802.11 standards.

A Wi Fi network allows devices such as laptops, smartphones, and tablets to communicate over a wireless LAN.

14. Infrastructure Mode

In infrastructure mode, wireless devices communicate through an access point. The access point acts as the central point of wireless communication and can connect wireless devices to a wired network.

15. Ad Hoc Mode

In ad hoc mode, wireless devices communicate directly with one another without requiring a central access point. It is useful for certain temporary or small wireless networks.

16. Wired LAN and Wireless LAN

Basis	Wired LAN	Wireless LAN
Connection	Uses physical cables.	Uses wireless radio communication.
Common Technology	Ethernet.	IEEE 802.11 and Wi Fi.
Mobility	Limited.	High.
Installation	Requires cable installation.	Usually requires less physical cabling.
Interference	Depends on cable type and environment.	Can be affected by wireless interference.
Performance	Generally stable and predictable.	Can vary with distance, obstacles, interference, and network conditions.
Security	Physical access to the cable may be required for direct access.	Requires appropriate wireless security because signals travel through the air.

17. LAN and WAN

A LAN covers a limited geographical area, while a WAN connects networks over large geographical distances.

Feature	LAN	WAN
Coverage	Small geographical area.	Large geographical area.
Typical Use	Home, office, laboratory, campus.	Connecting networks across cities, countries, or continents.
Ownership	Often managed by one organization.	May involve multiple organizations and service providers.
Communication	Usually high speed within the local network.	Performance depends on the underlying wide area connections.

18. MAC Address in a LAN

A MAC address is a link layer address associated with a network interface. Ethernet switches use MAC addresses to make forwarding decisions within a LAN.

The MAC address helps identify the intended local network interface for frame delivery.

19. Broadcast in a LAN

A broadcast is a transmission intended for all appropriate devices within a broadcast domain. Ethernet networks use a broadcast MAC address for broadcast frames.

Broadcast communication is useful for certain network operations, but excessive broadcast traffic can reduce network performance.

20. Collision Domain and Broadcast Domain

20.1 Collision Domain

A collision domain is a network area in which devices can potentially compete for the same shared transmission medium and experience collisions. Modern switched full duplex Ethernet networks greatly reduce this issue.

20.2 Broadcast Domain

A broadcast domain is the set of devices that receive a broadcast frame at the Data Link Layer. Routers separate broadcast domains.

21. Advantages of LAN

1. Resource sharing.
2. File and data sharing.
3. Printer and device sharing.
4. Fast communication between connected devices.
5. Centralized management of network resources.
6. Shared Internet access.

22. Limitations of LAN

1. A LAN has limited geographical coverage.
2. Network installation and maintenance require planning.
3. A failure of important network equipment can affect connected devices.
4. Security must be properly managed to prevent unauthorized access.

23. Simple Real Life Example

Consider a college computer laboratory containing thirty computers. If the computers are connected using Ethernet cables and a switch, it forms a wired LAN.

If students connect their laptops and smartphones to the same network using Wi Fi through an access point, the wireless part forms a wireless LAN.

24. Important Definitions to Learn

LAN: A network that connects devices within a limited geographical area.

Wired LAN: A LAN in which devices communicate through physical transmission media.

Wireless LAN: A LAN in which devices communicate using wireless transmission, usually radio waves.

Ethernet: A widely used LAN technology standardized under the IEEE 802.3 family.

IEEE 802.11: A family of IEEE standards for wireless local area networks.

Access Point: A device that provides wireless connectivity and connects wireless devices to a network.

Wi Fi: A technology used for wireless network connectivity based on IEEE 802.11 standards.

MAC Address: A link layer address associated with a network interface and used for local frame delivery.

Collision Domain: A network area in which devices can potentially compete for a shared transmission medium.

Broadcast Domain: A group of devices that can receive a broadcast at the Data Link Layer.

Topic 8: Virtual LAN

1. Introduction to VLAN

VLAN stands for Virtual Local Area Network. A VLAN is a logical network created within a physical network. It allows devices connected to the same physical switching infrastructure to be divided into separate logical networks.

A VLAN can group devices according to department, function, or requirement instead of only according to their physical location.

2. Why Do We Need VLAN?

In a traditional LAN, devices connected to the same switched network may belong to the same broadcast domain. As the number of devices increases, managing the network and controlling broadcast traffic can become difficult. VLANs are used to:

1. Divide a large LAN into smaller logical networks.
2. Separate users or devices according to their requirements.
3. Reduce the size of broadcast domains.
4. Improve network organization.
5. Improve security through logical separation.
6. Make network management more flexible.

3. Physical LAN and Virtual LAN

A physical LAN is based mainly on the physical connections between devices. A VLAN creates logical separation within the physical network.

Basis	Traditional Physical LAN	VLAN
Network Division	Mainly based on physical connections.	Based on logical grouping.
Broadcast Domain	Devices in the same broadcast domain receive broadcasts.	Different VLANs normally form separate broadcast domains.
Location	Devices are commonly grouped according to physical location.	Devices can belong to the same logical group even when physically separated.
Flexibility	Changing groups may require physical changes.	Logical group membership can be changed through configuration.
Management	Can become difficult as the network grows.	Provides more flexible logical organization.

4. VLAN and Switch

VLANs are commonly configured on managed network switches. A switch can assign ports to different VLANs.

For example, a switch may have some ports assigned to VLAN 10 for the Computer Science department and other ports assigned to VLAN 20 for the Administration department.

Even if the devices are connected to the same physical switch, devices in different VLANs are logically separated.

5. VLAN ID

A VLAN is identified using a VLAN ID. The VLAN ID allows network devices to distinguish traffic belonging to different VLANs.

For example, VLAN 10 and VLAN 20 represent two different VLANs. The exact VLAN numbers used in a network depend on the network configuration.

6. Access Port

An access port is a switch port that normally carries traffic belonging to a single VLAN. It is commonly used to connect end devices such as computers, printers, and other network hosts.

When a computer is connected to an access port, the computer normally does not need to understand VLAN tagging for basic access to that VLAN.

7. Trunk Port

A trunk port is a switch port that can carry traffic for multiple VLANs over the same physical connection. Trunk links are commonly used between switches and between a switch and a router or other networking device that needs to handle multiple VLANs.

8. VLAN Tagging

VLAN tagging is the process of adding information to an Ethernet frame to identify the VLAN to which the frame belongs.

IEEE 802.1Q is the commonly used standard for VLAN tagging in Ethernet networks.

9. IEEE 802.1Q

IEEE 802.1Q defines a method for identifying VLAN information in Ethernet frames. This makes it possible for multiple VLANs to share a trunk link.

The VLAN information added to the Ethernet frame is commonly called a VLAN tag.

10. Inter VLAN Communication

Devices in different VLANs are normally separated at the Data Link Layer and cannot communicate directly as if they were in the same VLAN.

If communication between different VLANs is required, a Layer 3 device such as a router or a Layer 3 switch can perform routing between the VLANs.

This process is called inter VLAN routing.

11. Broadcast Domain and VLAN

One of the most important features of VLANs is that each VLAN normally represents a separate broadcast domain.

A broadcast sent within one VLAN is normally not forwarded to devices belonging to another VLAN. This limits the spread of broadcast traffic.

12. Example of VLAN

Suppose a college has one large physical network containing computers from the Computer Science department, Administration department, and Library.

The network administrator can create separate VLANs for these groups. The physical network may remain the same, but the devices are logically separated.

Group	Example VLAN	Purpose
Computer Science	VLAN 10	Computer Science users and systems.
Administration	VLAN 20	Administrative users and systems.
Library	VLAN 30	Library users and systems.

In this example, a broadcast generated in VLAN 10 does not normally reach devices in VLAN 20 or VLAN 30.

13. Advantages of VLAN

1. Reduces broadcast domain size.
2. Improves logical network organization.
3. Provides logical separation between groups of devices.
4. Can improve security by separating different groups.
5. Allows users to be grouped according to function rather than only physical location.
6. Makes network changes easier because logical groups can be changed through configuration.

7. Can improve network management.

14. Limitations of VLAN

1. VLAN configuration requires managed network equipment and proper planning.
2. Incorrect VLAN configuration can cause communication problems.
3. Communication between VLANs requires Layer 3 routing when needed.
4. Troubleshooting can become more complex in large VLAN based networks.

15. VLAN and Security

VLANs can provide logical separation between groups of devices. **For example**, administrative systems can be placed in a different VLAN from student computers.

However, VLAN separation alone should not be considered complete security. Firewalls, access control, authentication, and other security mechanisms may also be required.

16. VLAN and Broadcast Traffic

Broadcast traffic is limited to the VLAN or broadcast domain in which it originates. Therefore, dividing a large LAN into multiple VLANs can reduce the number of devices that receive each broadcast.

This can help improve network performance and management, especially in larger networks.

17. VLAN Membership

A device can be assigned to a VLAN in different ways depending on the network design and switch capabilities. Common approaches include:

1. Port based VLAN: A switch port is assigned to a particular VLAN.
2. MAC based VLAN: VLAN membership can be associated with the MAC address of a device.
3. Other policy based methods may also be used in advanced network environments.

18. VLAN and Network Topology

VLAN is not a physical topology. Topology describes how devices are physically or logically connected, while VLAN describes how devices are logically grouped into separate networks.

A single physical switched topology can contain multiple VLANs.

19. VLAN and LAN

Basis	LAN	VLAN
Meaning	A network connecting devices within a limited geographical area.	A logical network created within a physical network.
Grouping	Often based on physical network connections.	Based on logical grouping.
Broadcast Domain	May contain many devices.	Each VLAN normally forms a separate broadcast domain.
Flexibility	Physical changes may be needed for some reorganizations.	Logical grouping can be changed through configuration.
Example	A college computer laboratory network.	A VLAN containing all faculty computers across different physical locations.

20. Access Port and Trunk Port

Feature	Access Port	Trunk Port
VLANs Carried	Normally one VLAN.	Multiple VLANs.
Typical Connection	End device such as a computer.	Switch to switch or network device connection.

VLAN Tagging	Frames sent to an end device are generally untagged in common access configurations.	VLAN identification is carried for multiple VLANs using tagging such as IEEE 802.1Q.
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21. Simple Real Life Example

Imagine a college with one large building and one physical network. Students, teachers, and administrative staff all use the same network infrastructure.

Instead of creating completely separate physical networks, the administrator can create separate VLANs for students, teachers, and administration.

The physical cables and switches can remain shared, while the logical networks remain separated.

22. Important Definitions to Learn

VLAN: A Virtual Local Area Network that logically divides a physical network into separate networks.

VLAN ID: An identifier used to distinguish one VLAN from another.

Access Port: A switch port that normally carries traffic belonging to one VLAN.

Trunk Port: A switch port that carries traffic for multiple VLANs.

VLAN Tagging: The process of adding VLAN identification information to an Ethernet frame.

IEEE 802.1Q: An IEEE standard that defines VLAN identification and tagging for Ethernet networks.

Broadcast Domain: A group of devices that can receive a broadcast at the Data Link Layer.

Inter VLAN Routing: Routing that allows communication between different VLANs using a Layer 3 device.

Unit 1: Data Communication Components

Topic 9: Multiplexing

1. Introduction to Multiplexing

Multiplexing is a technique used to transmit multiple signals through a single communication channel.

Instead of providing a separate communication channel for every signal, multiplexing allows several signals to share one channel. This makes better use of the available communication medium.

2. Why Is Multiplexing Needed?

Communication channels can have a large capacity, but a single signal may not always use the entire available capacity. Multiplexing allows multiple signals to share the same channel.

The main reasons for using multiplexing are:

1. Better utilization of available bandwidth.
2. Reduction in the number of physical communication links required.
3. More efficient use of expensive transmission media.
4. Ability to carry multiple signals over one communication channel.
5. Improved efficiency of communication systems.

3. Basic Concept of Multiplexing

A device called a multiplexer, or MUX, combines multiple input signals and sends them through a shared communication channel.

At the receiving end, a demultiplexer, or DEMUX, separates the combined signals and delivers them to their appropriate destinations.

Component	Function
Multiplexer	Combines multiple input signals into one shared signal stream.
Communication Channel	Carries the combined signal.
Demultiplexer	Separates the combined signal into the original signals.

4. Simple Example of Multiplexing

Suppose four users need to communicate with a receiver. Without multiplexing, four separate communication channels may be required.

With multiplexing, the four signals can share one communication channel. A multiplexer combines the signals before transmission, and a demultiplexer separates them at the receiving side.

5. Types of Multiplexing

The main multiplexing techniques covered in this topic are Frequency Division Multiplexing, Time Division Multiplexing, and Wavelength Division Multiplexing.

Technique	Basic Idea	Commonly Associated Medium
FDM	Different signals use different frequency bands.	Electrical or radio communication channels
TDM	Different signals use different time slots.	Digital communication systems
WDM	Different optical signals use different wavelengths of light.	Optical fiber

6. Frequency Division Multiplexing

Frequency Division Multiplexing, or FDM, divides the available frequency range of a communication channel into separate frequency bands.

Each signal is assigned a different frequency band. Multiple signals can therefore be transmitted simultaneously over the same physical medium.

7. Working of FDM

1. The available frequency range is divided into multiple frequency bands.
2. Each input signal is assigned a separate frequency band.
3. The multiplexer combines the signals.
4. All signals are transmitted through the same communication channel at the same time.
5. The receiver uses filters and a demultiplexer to separate the signals.

8. Guard Band in FDM

A guard band is a small unused frequency range placed between adjacent frequency bands in FDM. The purpose of the guard band is to reduce interference or overlap between neighboring channels.

9. Applications of FDM

1. Traditional telephone systems.
2. Radio broadcasting.
3. Television broadcasting.
4. Cable communication systems.

10. Advantages of FDM

1. Multiple signals can be transmitted simultaneously.
2. Signals can use the channel continuously.
3. Suitable for systems where different signals can be assigned different frequency ranges.
4. It can make efficient use of the available frequency spectrum.

11. Disadvantages of FDM

1. Guard bands reduce the amount of usable bandwidth.
2. The system requires suitable filtering and frequency management.
3. Interference between frequency bands can affect communication if the system is not properly designed.

12. Time Division Multiplexing

Time Division Multiplexing, or TDM, allows multiple signals to share a communication channel by assigning different time intervals to different signals.

Instead of dividing the channel by frequency, TDM divides access to the channel by time.

13. Working of TDM

1. The available communication time is divided into time slots.
2. Each input signal is assigned one or more time slots.
3. Signals are transmitted during their assigned time slots.
4. The receiver identifies the time slots and separates the signals.
5. The original signals are delivered to their appropriate destinations.

14. Time Frame in TDM

A group of time slots assigned to different input signals is called a frame in TDM.

For example, if four signals are given one time slot each, a TDM frame can contain four time slots, one for each signal.

15. Types of TDM

15.1 Synchronous TDM

In synchronous TDM, each input source is assigned a fixed time slot in every frame, even if the source has no data to send.

15.2 Statistical TDM

In statistical TDM, time slots are assigned dynamically to sources that have data to transmit. This can improve channel utilization when sources generate data at different times.

16. Advantages of TDM

1. Does not require separate frequency bands for different signals.
2. Suitable for digital communication systems.
3. Can efficiently share a channel among multiple users.
4. Statistical TDM can improve utilization when traffic is irregular.

17. Disadvantages of TDM

1. Synchronization between sender and receiver is required.
2. Synchronous TDM may waste time slots when a source has no data.
3. Delay can occur because a source may need to wait for its assigned time slot.

18. Wavelength Division Multiplexing

Wavelength Division Multiplexing, or WDM, is a multiplexing technique used mainly with optical fiber.

In WDM, multiple optical signals are transmitted through the same optical fiber using different wavelengths of light.

Different wavelengths act like separate channels, allowing several signals to share one optical fiber.

19. Working of WDM

1. Multiple optical signals are generated.
2. Each signal is assigned a different wavelength of light.
3. An optical multiplexer combines the wavelengths.
4. The combined optical signal travels through one optical fiber.
5. An optical demultiplexer separates the wavelengths at the receiving end.
6. The individual signals are delivered to their destinations.

20. Advantages of WDM

1. Provides very high capacity.
2. Allows multiple optical signals to share one fiber.
3. Makes efficient use of optical fiber bandwidth.
4. Supports expansion of communication capacity by adding additional wavelengths.

21. Disadvantages of WDM

1. Optical multiplexing equipment can be expensive.
2. The system requires accurate optical components.
3. Design and maintenance can be more complex than simpler communication systems.

22. FDM, TDM and WDM Comparison

Feature	FDM	TDM	WDM
Full Form	Frequency Division Multiplexing	Time Division Multiplexing	Wavelength Division Multiplexing
Division Based On	Frequency	Time	Wavelength
Typical Signal	Electrical or radio signals	Digital signals	Optical signals
Common Medium	Copper, radio, and other frequency based systems	Digital communication channels	Optical fiber
Simultaneous Use	Signals use different frequency bands at the same time.	Signals use different time slots.	Signals use different wavelengths at the same time.
Main Advantage	Allows simultaneous transmission using different frequencies.	Efficiently shares a channel in time.	Provides very high capacity over optical fiber.

23. Difference Between FDM and TDM

Basis	FDM	TDM
Division	Channel is divided into frequency bands.	Channel is divided into time slots.
Transmission	Multiple signals can transmit at the same time using different frequencies.	Signals transmit during their assigned time slots.
Guard Band	Generally required between frequency bands.	Not required for separating frequency bands.
Synchronization	Frequency separation is the main requirement.	Timing and synchronization are important.
Common Use	Radio and traditional frequency based systems.	Digital communication systems.

24. Difference Between TDM and WDM

Basis	TDM	WDM
Division	Time slots.	Wavelengths of light.
Medium	Digital communication channels.	Optical fiber.
Signal	Usually digital electrical data.	Optical signals.
Transmission	Signals share the channel at different times.	Signals share the fiber at different wavelengths.

25. Multiplexing and Bandwidth Utilization

Bandwidth utilization refers to how effectively the available capacity of a communication channel is used.

Multiplexing improves bandwidth utilization by allowing multiple signals to share the same communication medium instead of leaving large portions of the channel unused.

For example, FDM uses different frequency bands, TDM uses different time slots, and WDM uses different optical wavelengths to allow multiple signals to share one medium.

26. Multiplexer and Demultiplexer

Device	Function
Multiplexer	Combines multiple input signals into a single output stream for transmission over a shared channel.
Demultiplexer	Separates the combined signal into individual output signals at the receiving side.

27. Simple Real Life Example

Imagine a road that must carry vehicles belonging to many groups. One way to organize traffic is to assign different lanes. This is similar to FDM, where different signals use different frequency bands.

Another method is to allow different groups to use the same road at different times. This is similar to TDM, where different signals use different time slots.

In optical fiber, different colors of light can carry different signals through the same fiber. This is similar to WDM, where different wavelengths carry different signals.

28. Important Definitions to Learn

Multiplexing: A technique that allows multiple signals to share one communication channel.

Multiplexer: A device that combines multiple input signals into a shared signal stream.

Demultiplexer: A device that separates a combined signal into individual signals.

FDM: A multiplexing technique in which different signals use different frequency bands.

Guard Band: An unused frequency range placed between adjacent FDM channels to reduce interference.

TDM: A multiplexing technique in which different signals use different time slots.

Synchronous TDM: TDM in which each source is assigned a fixed time slot in every frame.

Statistical TDM: TDM in which time slots are assigned dynamically to sources that have data to send.

WDM: A multiplexing technique in which different optical signals use different wavelengths of light through the same fiber.

Topic 10: Spread Spectrum

1. Introduction to Spread Spectrum

Spread spectrum is a communication technique in which a signal is intentionally spread over a much wider range of frequencies than the minimum range required to transmit the original information.

The transmitted signal therefore occupies a wider frequency range. The receiver uses the appropriate spreading information to recover the original data.

Spread spectrum is mainly associated with wireless communication and is useful for improving resistance to interference and allowing multiple users or signals to share a communication environment.

2. Basic Idea of Spread Spectrum

In ordinary narrowband communication, a signal uses a relatively small frequency range. In spread spectrum communication, the signal is distributed over a wider frequency range.

The spreading is controlled using a spreading code or a related technique. At the receiver, the same or a related code is used to recover the original information.

3. Why Is Spread Spectrum Needed?

Spread spectrum can provide several useful characteristics:

1. Better resistance to interference.
2. Reduced effect of narrowband interference.
3. Improved privacy compared with simple narrowband transmission because the signal may appear less obvious to an unintended receiver.
4. Support for multiple users in some spread spectrum systems.
5. Greater robustness in certain wireless communication environments.

4. Main Characteristics of Spread Spectrum

1. The transmitted signal occupies a bandwidth much greater than the original information requires.
2. A spreading code or frequency pattern is used.
3. The receiver uses the appropriate information to despread the received signal.
4. The technique can improve resistance to certain types of interference.
5. Spread spectrum is commonly used in wireless communication systems.

5. Spreading and Despreading

5.1 Spreading

Spreading is the process of increasing the bandwidth occupied by the information signal using a spreading technique.

5.2 Despreading

Despreading is the process at the receiver in which the spread signal is processed using the appropriate spreading information to recover the original data.

6. Types of Spread Spectrum

Two important spread spectrum techniques are Frequency Hopping Spread Spectrum and Direct Sequence Spread Spectrum.

Technique	Basic Idea
FHSS	The carrier frequency changes or hops among a set of frequencies according to a predefined sequence.
DSSS	The original data is combined with a high rate spreading sequence, causing the transmitted signal to occupy a wider bandwidth.

7. Frequency Hopping Spread Spectrum

Frequency Hopping Spread Spectrum, or FHSS, spreads communication by rapidly changing the carrier frequency among a set of frequency channels.

The sequence of frequency changes is controlled by a predefined hopping pattern. The receiver follows the same hopping pattern to receive and reconstruct the transmitted information.

8. Working of FHSS

1. A set of possible carrier frequencies is selected.
2. A hopping sequence determines the order in which frequencies are used.
3. The transmitter sends data using one frequency for a short period.
4. The transmitter changes to another frequency according to the hopping sequence.
5. The process continues during communication.
6. The receiver follows the same hopping sequence and receives the signal.

9. Frequency Hopping

The change from one carrier frequency to another is called frequency hopping. Each frequency may be used for a short interval before the transmitter moves to the next frequency.

If interference affects one frequency, communication can continue by moving to another frequency, depending on the system design and interference conditions.

10. Advantages of FHSS

1. Provides resistance to narrowband interference.
2. A signal is not continuously transmitted on one frequency.
3. Can support multiple users using different hopping sequences in suitable systems.
4. Can provide robustness in some wireless environments.

11. Disadvantages of FHSS

1. The transmitter and receiver must remain synchronized.
2. The system requires accurate control of the hopping sequence.
3. Communication equipment must support rapid frequency changes.

12. Direct Sequence Spread Spectrum

Direct Sequence Spread Spectrum, or DSSS, spreads the original data by combining it with a high rate sequence called a spreading sequence or spreading code.

The spreading sequence contains more bits or chips than the original data contains bits. This causes the transmitted signal to occupy a wider frequency range.

13. Working of DSSS

1. The original data is represented as a sequence of bits.
2. A high rate spreading sequence is generated.
3. The data is combined with the spreading sequence.
4. The resulting signal occupies a wider bandwidth.
5. The signal is transmitted through the communication medium.
6. The receiver uses the appropriate spreading sequence to despread the signal and recover the original data.

14. Chip and Chip Rate

In DSSS, the bits of the spreading sequence are commonly called chips. The chip rate is the number of chips transmitted per second.

The chip rate is higher than the original data bit rate. This higher rate is one of the reasons the transmitted signal occupies a wider bandwidth.

15. Processing Gain

Processing gain describes the improvement obtained by spreading and despreading a signal, particularly in relation to resistance against certain interference.

In a simplified view, a larger spreading bandwidth compared with the original data bandwidth can provide greater processing gain.

16. Advantages of DSSS

1. Provides resistance to certain types of interference.
2. Can reduce the effect of narrowband interference on the received information.
3. Can support multiple users in systems that use different spreading codes.
4. Provides robustness in suitable wireless communication environments.

17. Disadvantages of DSSS

1. Requires a high rate spreading sequence.
2. The transmitter and receiver need suitable code synchronization.
3. Requires more bandwidth than the original narrowband information signal.

18. FHSS and DSSS Comparison

Basis	FHSS	DSSS
Full Form	Frequency Hopping Spread Spectrum	Direct Sequence Spread Spectrum
Main Technique	Changes carrier frequency according to a hopping sequence.	Combines data with a high rate spreading sequence.
Frequency Use	Uses different frequencies at different times.	Uses a wider frequency range continuously during transmission.
Synchronization	Hopping sequence synchronization is required.	Spreading code synchronization is required.
Interference	Can avoid or reduce the effect of interference on particular frequencies.	Can reduce the effect of narrowband interference through spreading and despreading.
Important Concept	Frequency hopping.	Spreading code and chips.

19. Spread Spectrum and Interference

One important advantage of spread spectrum is its ability to reduce the effect of certain interference. In FHSS, the signal changes frequency, so interference affecting one frequency may not affect the signal continuously. In DSSS, the signal is spread over a wider bandwidth. After despreading, the desired signal can be recovered while some narrowband interference is reduced relative to the desired signal.

20. Spread Spectrum and Security

Spread spectrum can make a signal less obvious to an unintended receiver and can provide some resistance to interception in suitable systems. However, spread spectrum by itself is not a complete security mechanism. Modern wireless networks use additional security mechanisms such as authentication and encryption when strong security is required.

21. Applications of Spread Spectrum

1. Wireless communication systems.
2. Certain mobile communication systems.
3. Satellite communication.
4. Wireless local area networking technologies.
5. Systems that require resistance to interference.
6. Some positioning and communication systems.

22. Spread Spectrum and Bandwidth

The key idea of spread spectrum is that the transmitted signal uses a bandwidth much greater than the minimum bandwidth required for the original information.

This additional bandwidth is used to obtain useful properties such as interference resistance and robustness.

23. Simple Real Life Example

Imagine two people communicating in a crowded room. Instead of speaking continuously at one fixed location, one person changes locations according to a known pattern while the other person follows the same pattern. This is similar to the basic idea of FHSS.

For DSSS, imagine a short message being represented using a much longer agreed pattern. The receiver knows the pattern and can use it to recover the original message.

24. Important Definitions to Learn

Spread Spectrum: A communication technique in which a signal is intentionally spread over a bandwidth much wider than the minimum required for the original information.

Spreading: The process of increasing the bandwidth occupied by an information signal using a spreading technique.

Despreading: The receiver process used to recover the original information from a spread signal.

FHSS: A spread spectrum technique in which the carrier frequency changes among multiple frequencies according to a hopping sequence.

Frequency Hopping: The process of changing the carrier frequency according to a predefined sequence.

DSSS: A spread spectrum technique in which data is combined with a high rate spreading sequence.

Spreading Code: A sequence used to spread the original information signal in a spread spectrum system.

Chip: A bit or element of the high rate spreading sequence used in DSSS.

Chip Rate: The number of chips transmitted per second.

Processing Gain: A measure related to the improvement obtained through spreading and despreading, especially against certain interference.